

Turbulence Analysis Report -

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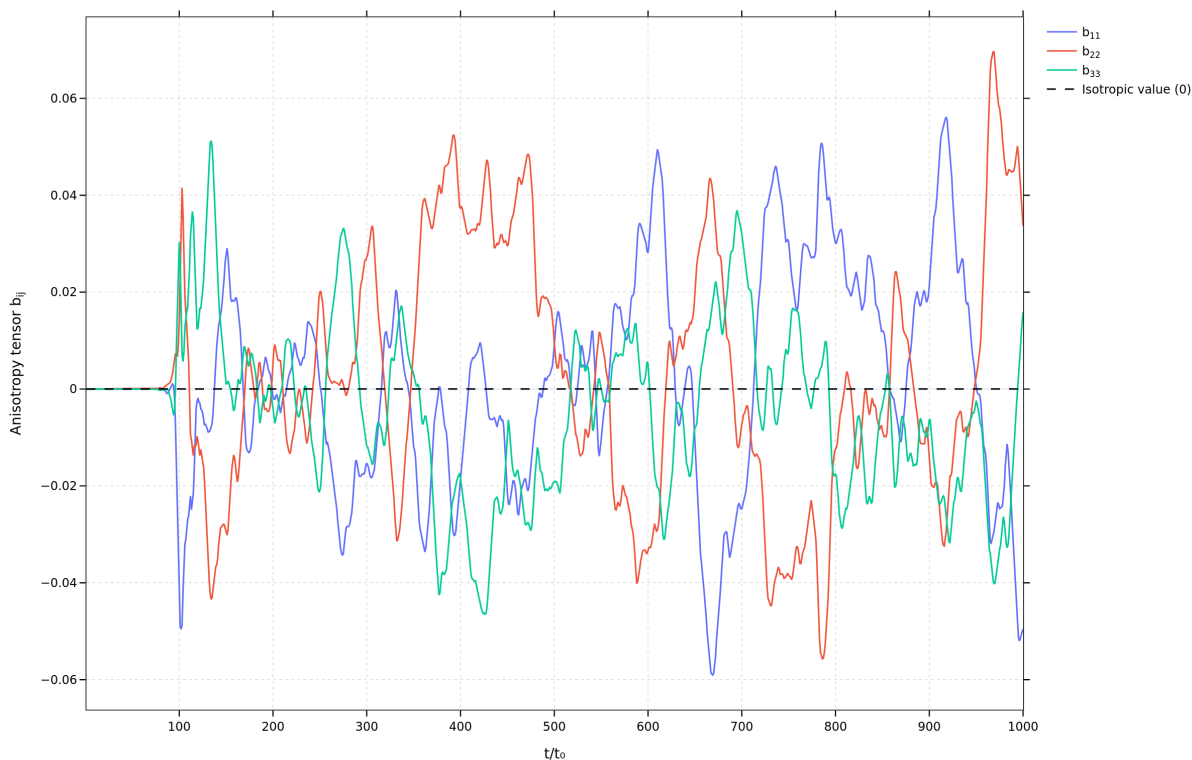


Figure 1. Plot of diagonal anisotropy components b_{11} , b_{22} , b_{33} as a function of t/t_0 , illustrating the evolution of anisotropy in the turbulence.

Theory of Diagonal Anisotropy

Diagonal Anisotropy Tensor Components (b_{ij})

The diagonal components of the anisotropy tensor, b_{11} , b_{22} , and b_{33} , quantify the departure from isotropic conditions along the principal axes. For isotropic turbulence, all diagonal components are zero. They are defined as:

$$b_{ii} = (R_{ii} - 2/3 k) / (2k)$$

where R_{ii} are the diagonal components of the Reynolds stress tensor and k is the turbulent kinetic energy.

Reference: Pope, S. B. (2001). Turbulent flows. Cambridge university press.

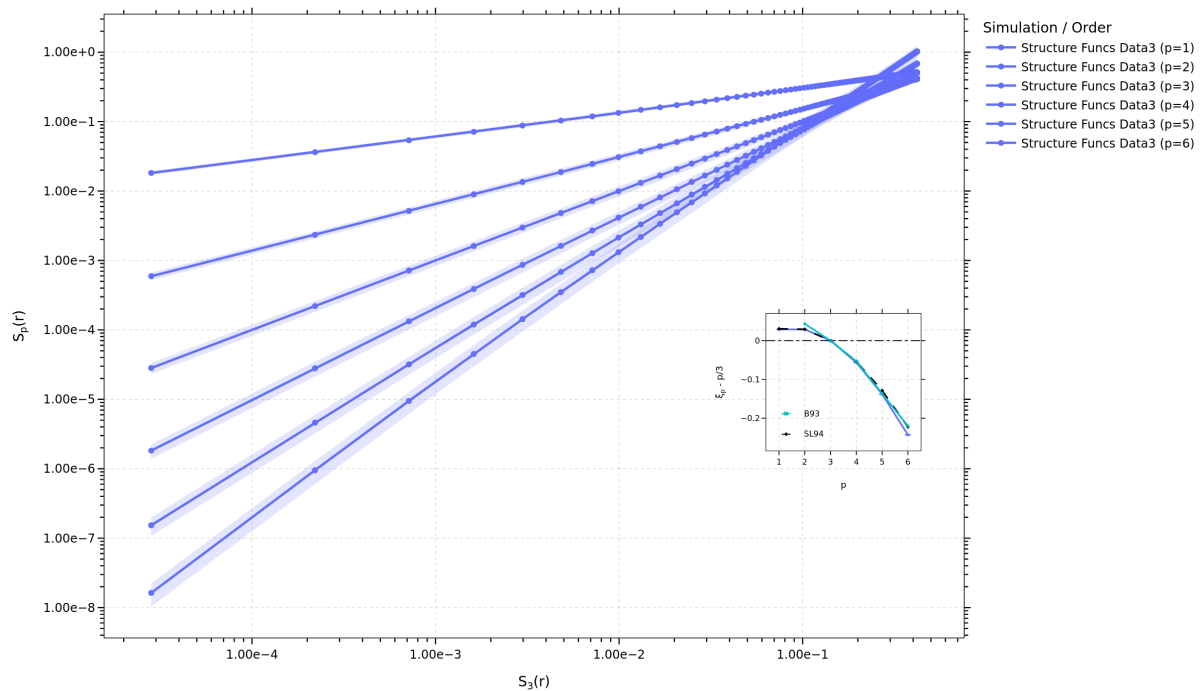


Figure 3. Plot of Extended Self-Similarity (ESS) for structure functions, demonstrating the scaling behavior of turbulence with an inset showing the scaling exponents.

Theory of Structure Functions and ESS

Include ESS equations, She-Leveque scaling, and references (Benzi et al. 1993, She & Leveque 1994).

Interpretation of Results

Provide a short prose explanation of the physical meaning of both figures, how they relate to isotropy and scaling in turbulence, and cite relevant literature.